

1st Year 1st Semester Topology Exam

August, 2023

Work the following problems and show all work. Support all statements to the best of your ability. Work each problem on a separate sheet of paper.

1. Let $f : X \rightarrow X$ be a map of a compact metric space to itself that satisfies the following condition: $d(f(x), f(y)) < d(x, y)$.
 - (a) Prove that f is continuous.
 - (b) Show that f has a fixed point and the fixed point is unique.
2. Let $X \subset \mathbb{R}^2$ be a countable subset. Prove that $\mathbb{R}^2 \setminus X$ is connected.
3. Let X be a connected normal space having more than one point. Can X be countable?
4. Let A be a proper subset of X and let B be a proper subset of Y . If X and Y are connected, show that $(X \times Y) - (A \times B)$ is connected.
5. Let I_O denote the square $[0, 1] \times [0, 1]$ given the lexicographic order topology. Is I_O
 - (a) compact?
 - (b) connected?
 - (c) path connected?

Answer the following with complete definitions or statements or short proofs.

6. Give definition of Lebesgue number.
7. Show that the power set $\mathcal{P}(X)$ has strictly greater cardinality than X .
8. Show that the 2-sphere is not homeomorphic to
 - (a) the circle S^1 ;
 - (b) the 2-plane $\mathbb{R}^2$.
9. Show that every compact Hausdorff space is regular.
10. Give definition of a quotient map. Let

$$Z = \{(x, y) \in \mathbb{R}^2 \mid xy = 1\} \cup (\{0\} \times \mathbb{R}) \subset \mathbb{R} \times \mathbb{R}$$

be given the subspace topology. Is the projection $f : Z \rightarrow \mathbb{R}$ to the first coordinate, $f(x, y) = x$, a quotient map?